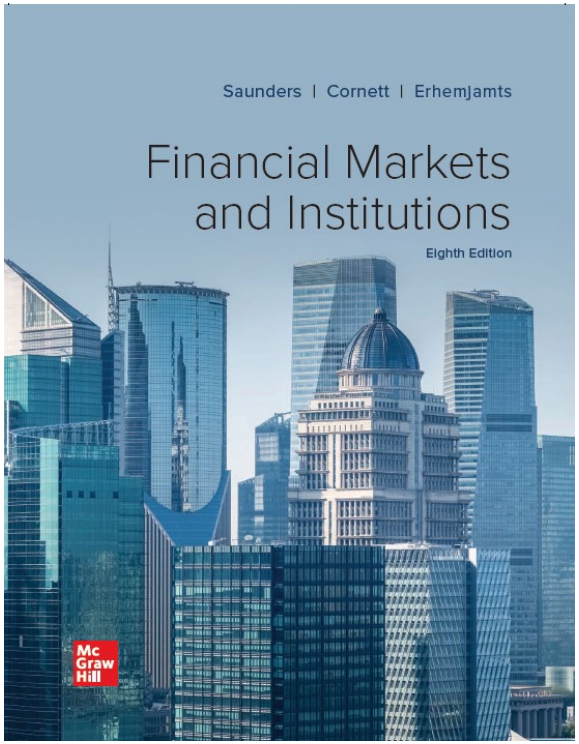
Chapter 6

Bond Markets



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Bond Markets

- Equity and debt instruments with maturities of more than one year trade in **capital markets**
- **Bonds** are long-term debt obligations issued by corporations and government units
  - Proceeds from a bond issue are used to raise funds to support long-term operations of the issuer
  - If the terms of the repayment are not met by the bond issuer, the bond holder (investor) has a claim on the assets of the bond issuer
- **Bond markets** are markets in which bonds are issued and traded
  - Treasury notes (T-notes) and bonds (T-bonds)
  - Municipal bonds
  - Corporate bonds

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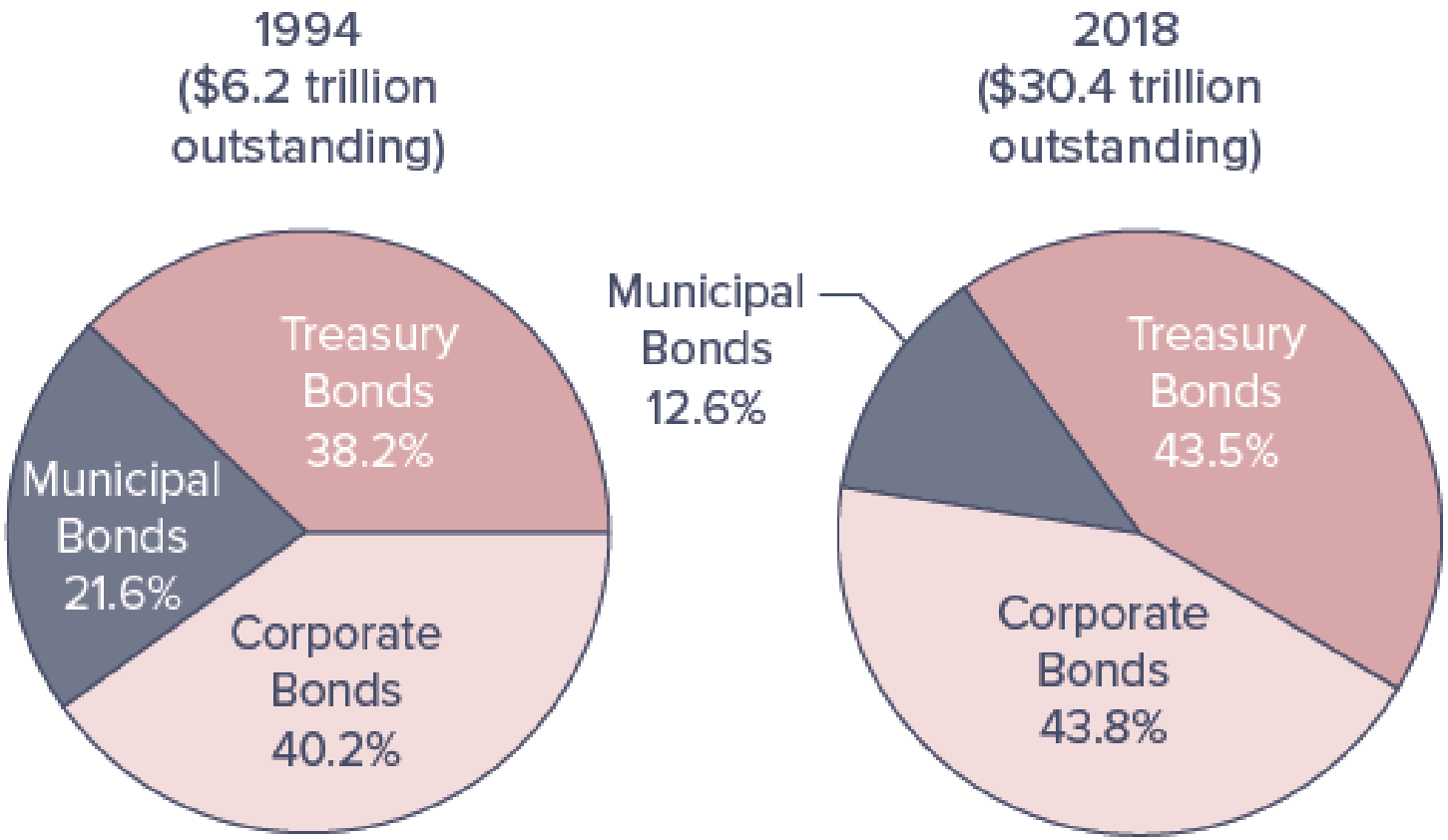
Bond Market Instruments Outstanding, 1994 - 2018



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Treasury Notes and Bonds

- **Treasury notes and bonds (T-notes and T-bonds)** are issued by the U.S. Treasury to finance the national debt and other government expenditures
  - $13.2 trillion outstanding in 2018
- The national debt (*ND*) reflects the historical accumulation of annual federal government deficits or expenditures (*G*) minus taxes (*T*) over the last 200-plus years, as follows:

$$ND_t = \sum_{t=1}^{N} (G_t - T_t)$$

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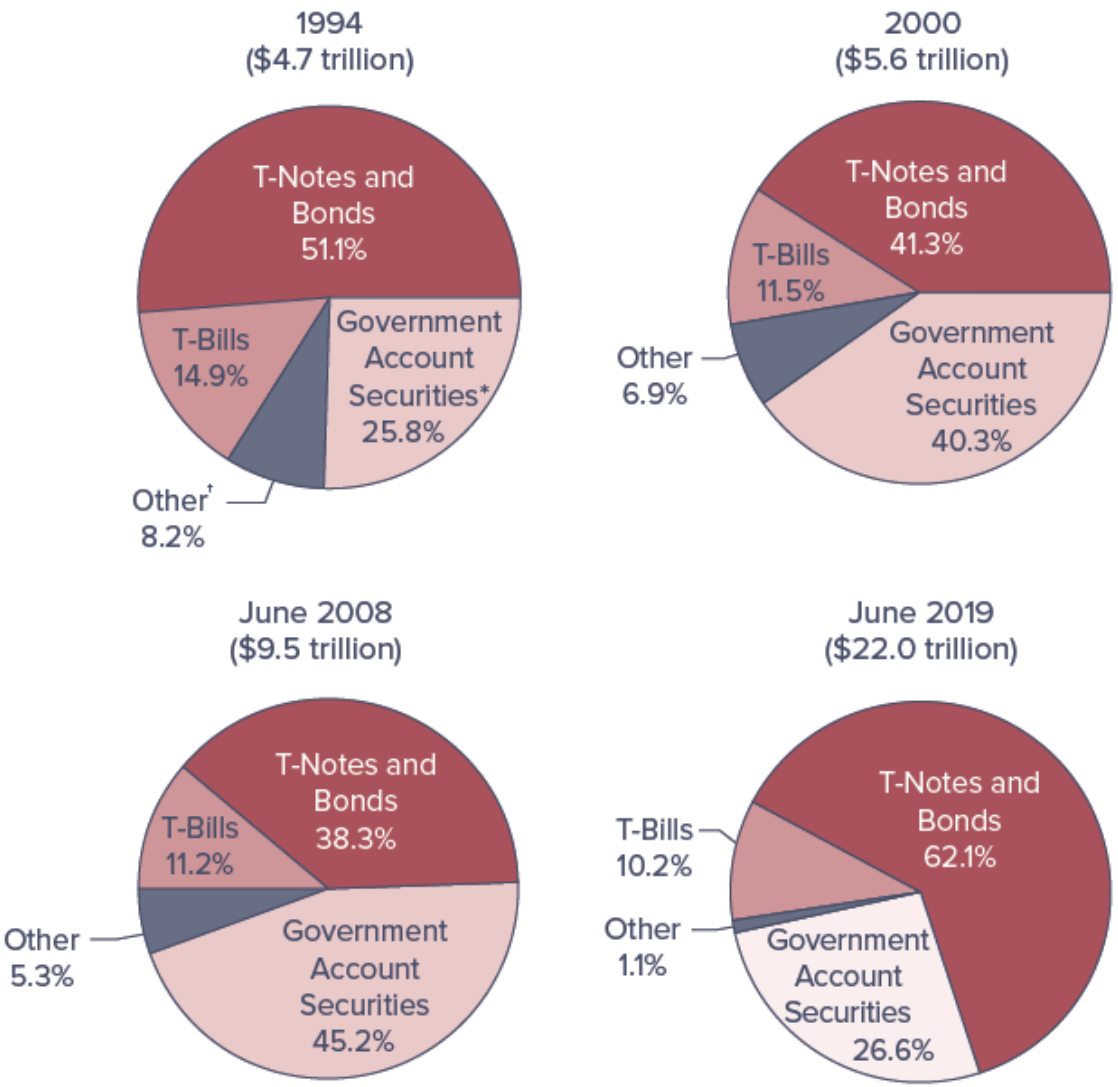
Composition of the U.S. National Debt



Treasury Notes and Bonds (Continued)

- T-notes and T-bonds are backed by the full faith and credit of the U.S. government and are, therefore, default risk free
- T-notes and T-bonds pay relatively low rates of interest (yields to maturity) to investors
- T-notes and T-bonds pay coupon interest (semiannually)
- T-bills have an original maturity of one year or less, T-notes have original maturities from over 1 to 10 years, while T-bonds have original maturities from over 10 years
- Like T-bills, once issued T-notes and T-bonds trade in very active secondary markets

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Treasury Notes and Bonds (Concluded)

- Treasury issued two types of notes and bonds: fixed principal and inflation-indexed
  - Both types pay interest twice per year
  - Principal value used to determine the percentage interest payment (coupon) on inflation-indexed bonds is adjusted to reflect inflation (measured by the CPI)
    - In other words, the semiannual coupon payments and the final principal payment are based on the inflation-adjusted principal value of the security
- Prices are quoted as percentages of the face value on the Treasury security, while coupon rates are set at intervals of 0.125 (or 1/8 of 1) percent

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STRIPS

- **Separate Trading of Registered Interest and Principal Securities (STRIPS)** is a Treasury security in which the periodic interest payment is separated from the final principal payment
  - Each of the components of the STRIPS are often referred to as “Treasury zero bonds” or “Treasury zero-coupon bonds” because investors in the individual components receive only the single stripped payments in which they invest
- Created by the U.S. Treasury in response to the separate trading of Treasury security principal and interest that had been developed by securities firms
- May be used to immunize against interest rate risk

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Treasury Note and Bond Yields

- Bond value calculated from the following formula is the price (referred to as the clean price) quoted in the financial press:

$$V_b = \frac{INT}{m}\sum_{t=1}^{mN}\left(\frac{1}{1+\frac{r_b}{m}}\right)^t + \frac{M}{\left(1+\frac{r_b}{m}\right)^{mN}}$$

$$= \frac{INT}{m}\left[\frac{1-\frac{1}{\left(1+\frac{r_b}{m}\right)^{mN}}}{\frac{r_b}{m}}\right] + M\left[\frac{1}{\left(1+\frac{r_b}{m}\right)^{mN}}\right]$$

where

V_b = Present value of the bond
M = Par or face value of the bond
INT = Annual interest (or coupon) payment on the bond equals the par value times the coupon rate
N = Number of years until the bond matures
m = Number of times per year interest is paid
r_b = Interest rate used to discount cash flows on the bond

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Accrued Interest

- **Accrued interest** is the portion of the coupon payment accrued between the last coupon payment and the settlement day
  - Paid by the buyer to the seller if the T-note or T-bond is purchased between interest payment dates

- The purchase price of the T-bond or T-note plus accrued interest is called the *full price* or the *dirty price* of the security, while the price without accounting for accrued interest is the *clean price*

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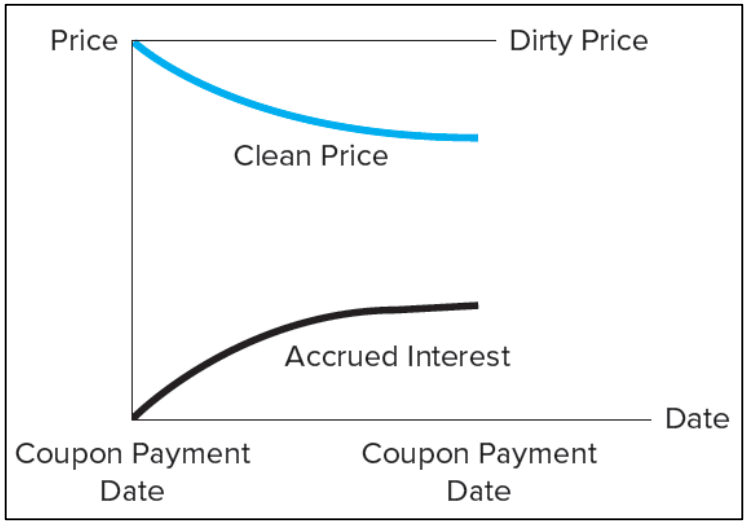
Accrued Interest (Continued)

- Accrued interest on a T-note or T-bond is based on the actual number of days the bond was held by the seller since the last coupon payment:

$$\text{Accrued interest} = \frac{INT}{2} \times \frac{\text{Actual number of days since last coupon payment}}{\text{Actual number of days in coupon period}}$$



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Calculation of Accrued Interest and Yield to Maturity on a Bond

On August 5, 2019, you purchase a $10,000 T-note that matures on May 15, 2025 (settlement occurs one day after purchase, so you receive actual ownership of the bond on August 6, 2019). The coupon rate on the T-note is 5.875 percent and the current price quoted on the bond in the financial press is 101.3437. The last coupon payment occurred on May 15, 2019 (82 days before settlement), and the next coupon payment will be paid on November 15, 2019 (102 days from settlement). We illustrate this time line in Figure 6–4.

The accrued interest due to the seller from the buyer at settlement is calculated as:

$$(5.875\%/2) \times 82/184 = 1.3251\%$$

of the face value of the bond, or $130.91. The dirty price of this transaction, or the full price paid by the bond buyer, is:

$$\text{Clean price} + \text{Accrued interest} = \text{Dirty price}$$
$$101.3437\% + 1.3091\% = 102.6528\%$$

of the face value of the bond, or $10,265.28 per $10,000 face value bond.

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Treasury Inflation Protected Securities (TIPS)

- In January 1997, the U.S. Treasury began issuing inflation-indexed bonds called Treasury Inflation Protected Securities (TIPS), which provide returns tied to the inflation rate
  - Outstanding value of TIPS was over $1.4 trillion by June 2019
- Unlike the fixed-principal bonds, the principal value of a TIPS bond can increase (or decrease) every six months by the amount of U.S. inflation (or deflation) as measured by the percentage change in the CPI
- Used by investors who wish to earn a rate of return on their investments that keeps up with the inflation rate over time

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Primary and Secondary Market Trading in T-Notes and T-Bonds

- U.S. Treasury sells T-notes and T-bonds through competitive and noncompetitive Treasury auctions
  - Auction is a single-price auction—all bidders pay the same price, which is the price associated with the highest of the competitive yields bid
  - At each auction, noncompetitive bids are filled first
  - Next, competitive bids are ranked from the lowest to highest yield (a bidder willing to accept the lowest yield is willing to pay the highest price)
  - Coupon rate of the auctioned notes or bonds is the stop-out yield rounded down to the nearest 1/8 percent
- Most secondary trading occurs directly through broker and dealer trades

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Municipal Bonds

- **Municipal bonds** are securities issued by state and local governments; primary reasons for issuances are the following:
  - Fund imbalances between expenditures and receipts
  - Finance long-term capital outlays
- Attractive to household investors because interest is exempt from federal and most state/local income taxes
- **General obligation (GO) bonds** are backed by the full faith and credit of the issuer
- **Revenue bonds** are sold to finance specific revenue-generating project and are backed by the cash flows from that project

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Municipal Bond Yields

- The after-tax (or equivalent tax-exempt) yield on a taxable bond can be calculated as follows:

$$r_a = r_b(1 - t)$$

- The yield on a tax-exempt municipal bond can be calculated as follows:

$$r_b = r_a/(1 - t)$$

where

r_a = After-tax (equivalent tax-exempt) yield on a taxable corporate bond
r_b = Before-tax yield on a taxable bond
t = Marginal income tax rate of the bond holder (i.e., the sum of his or her marginal federal, state, and local taxes)[8]

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Primary Market Trading

- **Firm commitment underwriting** is the issue of securities by an investment bank in which the investment bank guarantees the issuer a price for newly issued securities by buying the whole issue at a fixed price from the issuer
  - The investment bank then seeks to resell these securities to suppliers of funds (investors) at a higher price
- **Best efforts offering** is the issue of securities in which the investment bank does not guarantee a price to the issuer and acts more as a placing or distribution agent on a fee basis related to its success in placing the issue
- **Private placement** is a security issue placed with one or a few large institutional buyers

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Secondary Market Trading

- Secondary market trading for municipal bonds is relatively infrequent
- Thin trading is mainly a result of a lack of information on bond issuers, as well as special features (such as covenants) that are built into those bonds' contracts
  - Information on municipal bond issuers (particularly of smaller government units) is generally more costly to obtain and evaluate, although this is in part offset by bond rating agencies

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Corporate Bonds

- **Corporate bonds** are long-term obligations issued by corporations
  - Minimum denomination on publicly traded corporate bonds is $1,000
  - Coupon-paying corporate bonds generally pay interest semiannually
- A **bond indenture** is the legal contract that specifies the rights and obligations of the bond issuer and the bond holders
  - Contains several covenants associated with a bond issue
  - Bond covenants describe rules and restrictions placed on the bond issuer and bond holders

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Corporate Bonds Characteristics

- Bearer versus Registered Bonds
  - **Bearer bonds** have coupons attached to the bonds, and the holder presents the coupons to the issuer for payments of interest when they come due
  - **Registered bonds** are those in which the owner is recorded by the issuer and the coupon payments are mailed to the registered owner
- Term versus Serial Bonds
  - **Term bonds** are those in which the entire issue matures on a single date
  - **Serial bonds** mature on a series of dates, with a portion of the issue paid off on each

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Corporate Bonds Characteristics (Continued)

- **Mortgage bonds** are issued to finance specific projects, which are pledged as collateral for the bond issue
- Debentures and Subordinated Debentures
  - **Debentures** are bonds backed solely by the general credit worthiness of the issuing firm, unsecured by specific assets or collateral
  - **Subordinated debentures** are bonds that are unsecured and are junior in their rights to mortgage bonds and regular debentures
- **Convertible bonds** may be exchanged for another security of the issuing firm at the discretion of the bond holder

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Corporate Bonds Characteristics (Concluded)

- Bonds issued with **stock warrants** give the bond holder the option to detach the warrants to purchase common stock at a prespecified price up to a prespecified date
- Callable bonds
  - Many corporate bond issues include a **call provision**, which allows the issuer to require the bond holder to sell the bond back to the issuer at a given (call) price— usually set above the par value of the bond
  - The difference between the call price and the face value on the bond is the **call premium**
- **Sinking fund provision** is a requirement that the issuer retire a certain amount of the bond issue each year

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The Trading Process for Corporate Bonds

- Primary sales of corporate bond issues occur through either a public sale (issue) or a private placement in a manner identical to that discussed for municipal bonds
- Two secondary markets trade corporate bonds:
  1. Exchange market (e.g., NYSE Bonds)
  2. Over-the-counter (OTC) market
- Major exchange for corporate bonds is the New York Stock Exchange Bonds Trading Platform
  - Most of the trading on the NYSE's bond market is completed through its NYSE Arca all-electronic trading system
- Only a small amount of bond trading volume occurs on organized exchanges
  - Most bonds are traded OTC among major bond dealers

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Bond Ratings and Interest Rate Spreads

- Three major bond rating agencies are the following:
  - Moody's
  - Standard & Poor's (S&P)
  - Fitch Ratings
- Each of these companies rank bonds based on the perceived probability of issuer default and assign a rating based on a letter grade
  - The highest credit quality (lowest default risk) that rating agencies assign is a triple-A (Aaa for Moody's and AAA for S&P and Fitch)
  - Bonds rated below Baa by Moody's and BBB by S&P and Fitch are considered to be speculative grade bonds and are often termed **junk bonds**, or high-yield bonds

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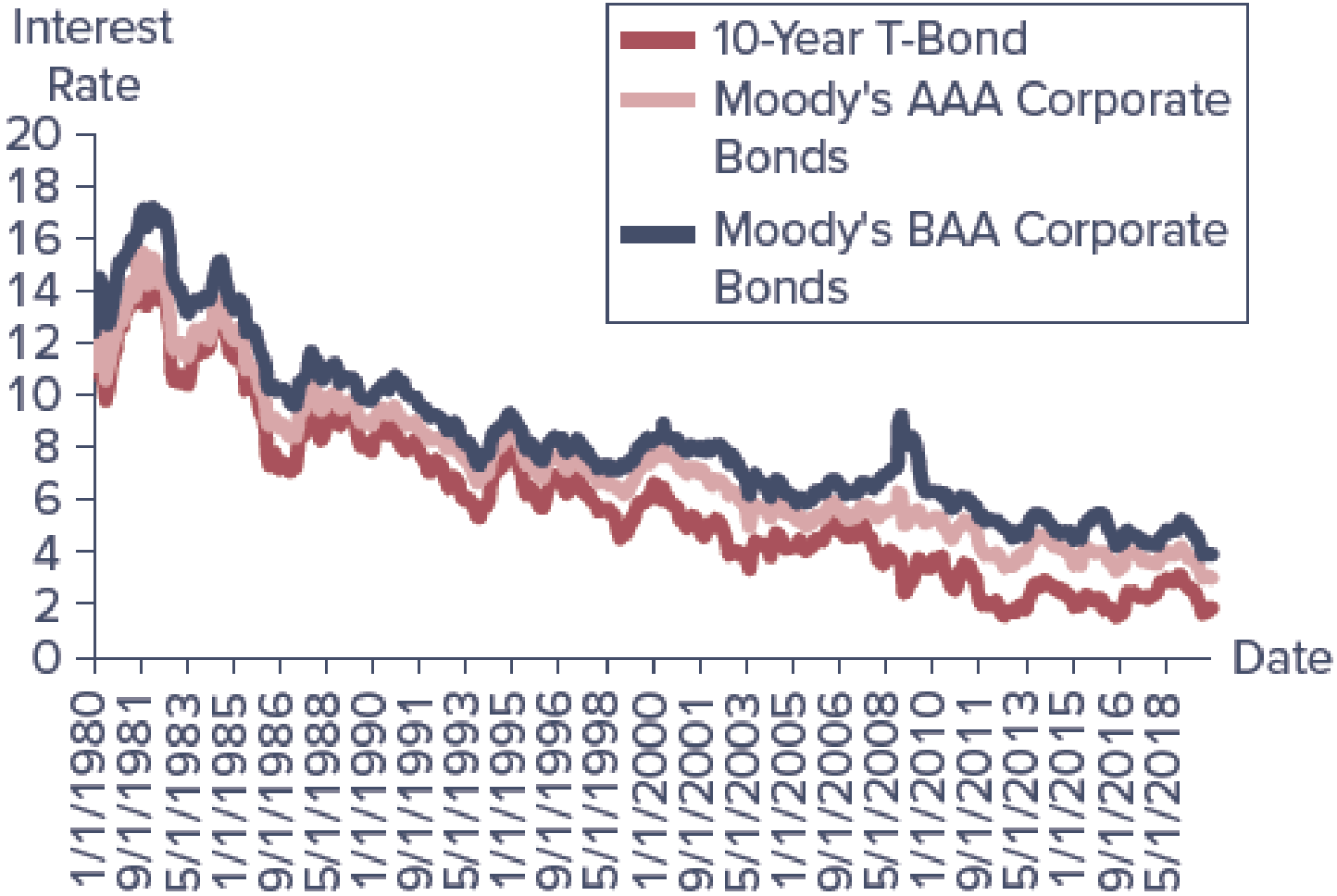
Bond Credit Ratings

Explanation	Moody's	S&P	Fitch
Best quality; smallest degree of risk	Aaa	AAA	AAA
High quality; slightly more long-term risk than top rating	Aa1	AA+	AA+
	Aa2	AA	AA
	Aa3	AA–	AA–
Upper medium grade; possible impairment in the future	A1	A+	A+
	A2	A	A
	A3	A–	A–
Medium grade; lacks outstanding investment characteristics	Baa1	BBB+	BBB+
	Baa2	BBB	BBB
	Baa3	BBB–	BBB–
Speculative issues; protection may be very moderate	Ba1	BB+	BB+
	Ba2	BB	BB
	Ba3	BB–	BB–
Very speculative; may have small assurance of interest and principal payments	B1	B+	B+
	B2	B	B
	B3	B–	B–
Issues in poor standing; may be in default	Caa	CCC	CCC
Speculative in a high degree; with marked shortcomings	Ca	CC	CC
Lowest quality; poor prospects of attaining real investment standing	C	C	C
Payment default		D	D

Sources: Websites for Moody's, Standard & Poor's, and Fitch Ratings, www.moodys.com; www.standardandpoors.com; www.fitchratings.com

Rates on Treasury Bonds



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Bond Market Indexes

- Indexes are managed by major investment banks and reflect both the monthly capital gain and loss on bonds in the index plus any interest (coupon) income earned

- Changes in values of broad bond market indexes can be used by bond traders to evaluate changes in the investment attractiveness of bonds of different types and maturities

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